stevecrates489-commits / tokcalc
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Commit 7517e36

oldni9 committed 1 minute ago · ✕ 0 / 1

feat(mcp): v0.2.0 — public hosted endpoint at /api/mcp

main

1 parent 00b6eef commit 7517e36

6 files changed

+13,658 -32

mini-services/mcp-server/package.json +8 -2

```
@@ -1,7 +1,7 @@
 {
   "name": "@tokcalc/mcp-server",
-  "version": "0.2.0-beta.1",
-  "description": "tokcalc MCP server — open-source LLM serving capacity planner for AI agents. v0.2.0-beta.1: stdio + stateless HTTP transport with bearer API key auth + KV-backed rate limiting. 7 read-only tools: estimate_capacity, compare_gpus, recommend_topology, estimate_api_vs_self_host, list_models, list_gpus, get_mlperf_benchmarks.",
+  "version": "0.2.0",
+  "description": "tokcalc MCP server — open-source LLM serving capacity planner for AI agents. v0.2.0: stdio + stateless HTTP transport with bearer API key auth + KV-backed rate limiting + public hosted endpoint. 7 read-only tools: estimate_capacity, compare_gpus, recommend_topology, estimate_api_vs_self_host, list_models, list_gpus, get_mlperf_benchmarks.",
   "type": "module",
   "license": "Apache-2.0",
   "bin": {
@@ -34,6 +34,12 @@
   "engines": {
     "node": ">=18"
   },
+  "exports": {
+    ".": "./dist/server.js",
+    "./server": "./dist/server.js",
+    "./server.js": "./dist/server.js",
+    "./package.json": "./package.json"
+  },
   "dependencies": {
     "@modelcontextprotocol/sdk": "^1.30.1",
     "@upstash/ratelimit": "^2.0.3",
```

mini-services/mcp-server/server.ts +1 -1

```
@@ -495,7 +495,7 @@ const TOOL_DEFINITIONS = [
  */
 export function createMcpServer(): Server {
   const server = new Server(
-    { name: "tokcalc", version: "0.2.0-beta.1" },
+    { name: "tokcalc", version: "0.2.0" },
     {
       capabilities: {
         tools: {},
```

package-lock.json +13,217

Load diff

Some generated files are not rendered by default. Learn more about customizing how changed files appear on GitHub.

package.json +8 -5

```
@@ -1,8 +1,8 @@
 {
-  "name": "@tokcalc/mcp-server",
-  "version": "0.1.3",
-  "mcpName": "io.github.stevecrates489-commits/tokcalc",
-  "private": false,
+  "name": "tokcalc",
+  "version": "0.3.0",
+  "private": true,
   "scripts": {
     "dev": "next dev -p 3000 2>&1 | tee dev.log",
     "build": "next build",
@@ -21,6 +20,7 @@
     "@dnd-kit/utilities": "^3.2.2",
     "@hookform/resolvers": "^5.1.1",
     "@mdxeditor/editor": "^3.39.1",
+    "@modelcontextprotocol/sdk": "^1.30.1",
     "@prisma/client": "^6.11.1",
     "@radix-ui/react-accordion": "^1.2.11",
     "@radix-ui/react-alert-dialog": "^1.1.14",
@@ -52,6 +52,9 @@
     "@reactuses/core": "^6.0.5",
     "@tanstack/react-query": "^5.82.0",
     "@tanstack/react-table": "^8.21.3",
+    "@tokcalc/mcp-server": "file:./mini-services/mcp-server",
+    "@upstash/ratelimit": "^2.0.3",
+    "@upstash/redis": "^1.34.0",
     "class-variance-authority": "^0.7.1",
     "clsx": "^2.1.1",
     "cmdk": "^1.1.1",
@@ -94,4 +97,4 @@
     "tw-animate-css": "^1.3.5",
     "typescript": "^5"
   }
-}
+}
```

src/app/mcp/page.tsx +71 -24

```
@@ -34,6 +34,15 @@ export default function McpDocsPage() {
     setTimeout(() => setCopied(null), 1500);
   };
 
+  const cursorHostedConfig = `{
+  "mcpServers": {
+    "tokcalc": {
+      "url": "https://tokcalc.vercel.app/api/mcp",
+      "headers": { "Authorization": "Bearer <your-api-key>" }
+    }
+  }
+}`;
+
   const cursorConfig = `{
   "mcpServers": {
     "tokcalc": {
@@ -70,7 +79,7 @@ PORT=8080 tokcalc-mcp-http`;
       <div className="mb-8">
         <div className="flex items-center gap-3 mb-3">
           <h1 className="text-3xl font-bold tracking-tight">tokcalc MCP Server</h1>
-          <Badge className="bg-emerald-500/10 text-emerald-600 border-emerald-500/30 text-[10px]">v0.2.0-alpha.1</Badge>
+          <Badge className="bg-emerald-500/10 text-emerald-600 border-emerald-500/30 text-[10px]">v0.2.0</Badge>
         </div>
         <p className="text-muted-foreground text-sm sm:text-base leading-relaxed">
           7 read-only planning tools for AI agents (Cursor, Claude Desktop, Cline). Ask your AI:
@@ -147,16 +156,60 @@ PORT=8080 tokcalc-mcp-http`;
         </CardContent>
       </Card>
 
-      {/* Install — HTTP (alpha) */}
-      <Card className="border-border/60 shadow-sm mb-8">
+      {/* Install — Public hosted endpoint (v0.2.0 final) */}
+      <Card className="border-emerald-500/30 shadow-sm mb-8">
         <CardHeader>
           <CardTitle className="text-base flex items-center gap-2">
             <Zap className="size-4 text-emerald-500" />
-            Self-hosted HTTP transport (v0.2.0-alpha.1)
+            Use the public hosted endpoint (v0.2.0 — recommended)
           </CardTitle>
           <CardDescription className="text-xs">
-            For agent platforms that need HTTP transport (no local <code>npx</code>). Stateless mode —
-            no sessions, no SSE stream, no auth in alpha.1.
+            No <code>npx</code> install needed — point your AI agent directly at the public URL.
+            Backed by bearer API key auth + Upstash Redis rate limiting.
+          </CardDescription>
+        </CardHeader>
+        <CardContent className="space-y-3">
+          <p className="text-xs text-muted-foreground">
+            <strong className="text-foreground">Public endpoint:</strong>{" "}
+            <code className="font-mono text-emerald-500">https://tokcalc.vercel.app/api/mcp</code>
+          </p>
+          <p className="text-xs text-muted-foreground">
+            Add to your Cursor / Claude Desktop config:
+          </p>
+          <div className="relative">
+            <pre className="bg-muted rounded-md p-3 text-xs font-mono overflow-x-auto border border-border/60"><code>{cursorHostedConfig}</code></pre>
+            <Button
+              size="sm"
+              variant="ghost"
+              className="absolute top-2 right-2 text-[10px]"
+              onClick={() => copyToClipboard(cursorHostedConfig, "cursor-hosted")}
+            >
+              {copied === "cursor-hosted" ? "✓ Copied" : "Copy"}
+            </Button>
+          </div>
+          <p className="text-xs text-muted-foreground mt-3">
+            <strong className="text-foreground">Get an API key:</strong> Email{" "}
+            <a href="mailto:hello@tokcalc.vercel.app?subject=MCP%20API%20key%20request" className="text-emerald-500 hover:underline">hello@tokcalc.vercel.app</a>{" "}
+            with subject "MCP API key request" and we&apos;ll send you a key within 24 hours.
+          </p>
+          <div className="text-[11px] text-emerald-600 dark:text-emerald-400 flex items-start gap-1.5 mt-2">
+            <span className="font-medium">✓ Production-safe:</span>
+            <span>v0.2.0 is bearer-authenticated (constant-time comparison) and rate-limited
+            (30/min per IP anonymous, 120/min per API key). Safe for public exposure.</span>
+          </div>
+        </CardContent>
+      </Card>
+
+      {/* Install — Self-hosted HTTP (for power users) */}
+      <Card className="border-border/60 shadow-sm mb-8">
+        <CardHeader>
+          <CardTitle className="text-base flex items-center gap-2">
+            <Terminal className="size-4 text-emerald-500" />
+            Self-host the HTTP server (advanced)
+          </CardTitle>
+          <CardDescription className="text-xs">
+            For air-gapped deployments, custom rate limits, or running on your own infra.
+            Same code as the public endpoint — just on your own server.
           </CardDescription>
         </CardHeader>
         <CardContent className="space-y-3">
@@ -185,10 +238,11 @@ PORT=8080 tokcalc-mcp-http`;
               {copied === "curl" ? "✓ Copied" : "Copy"}
             </Button>
           </div>
-          <div className="text-[11px] text-amber-600 dark:text-amber-400 flex items-start gap-1.5 mt-2">
-            <span className="font-medium">⚠ Alpha scope:</span>
-            <span>v0.2.0-alpha.1 has NO authentication or rate limiting. Do NOT expose to the public internet.
-            Bearer API key + KV-backed rate limits arrive in v0.2.0-beta.1.</span>
+          <div className="text-[11px] text-emerald-600 dark:text-emerald-400 flex items-start gap-1.5 mt-2">
+            <span className="font-medium">✓ Self-hosted:</span>
+            <span>v0.2.0 includes bearer API key auth + Upstash Redis rate limiting.
+            Set <code>MCP_API_KEY</code>, <code>UPSTASH_REDIS_REST_URL</code>, and <code>UPSTASH_REDIS_REST_TOKEN</code>
+            env vars before deploying to your own server.</span>
           </div>
         </CardContent>
       </Card>
@@ -230,24 +284,17 @@ PORT=8080 tokcalc-mcp-http`;
             </div>
           </div>
           <div className="flex items-start gap-3">
-            <Badge className="bg-amber-500/10 text-amber-600 border-amber-500/30 text-[9px] shrink-0">alpha</Badge>
-            <div>
-              <div className="font-medium">v0.2.0-alpha.1 — stateless HTTP transport</div>
-              <div className="text-muted-foreground">For agent platforms that need HTTP. No auth yet — local testing only.</div>
-            </div>
-          </div>
-          <div className="flex items-start gap-3">
-            <Badge variant="outline" className="text-[9px] shrink-0">next</Badge>
+            <Badge className="bg-emerald-500/10 text-emerald-600 border-emerald-500/30 text-[9px] shrink-0">✓ shipped</Badge>
             <div>
-              <div className="font-medium">v0.2.0-beta.1 — bearer API key + rate limiting</div>
-              <div className="text-muted-foreground">Public deploy-safe. KV-backed rate limits per IP / per key.</div>
+              <div className="font-medium">v0.2.0-alpha.1 + beta.1 — HTTP transport + auth + rate limit</div>
+              <div className="text-muted-foreground">Stateless Streamable HTTP + bearer API key + Upstash Redis rate limiting</div>
             </div>
           </div>
           <div className="flex items-start gap-3">
-            <Badge variant="outline" className="text-[9px] shrink-0">v0.2.0</Badge>
+            <Badge className="bg-emerald-500/10 text-emerald-600 border-emerald-500/30 text-[9px] shrink-0">✓ shipped</Badge>
             <div>
               <div className="font-medium">v0.2.0 — public hosted endpoint</div>
-              <div className="text-muted-foreground">mcp.tokcalc.app — no install needed for hosted clients</div>
+              <div className="text-muted-foreground">Live at tokcalc.vercel.app/api/mcp — no install needed</div>
             </div>
           </div>
           <div className="flex items-start gap-3">
@@ -310,8 +357,8 @@ PORT=8080 tokcalc-mcp-http`;
             applicationCategory: "DeveloperApplication",
             operatingSystem: "Cross-platform (Node.js 18+)",
             offers: { "@type": "Offer", price: "0", priceCurrency: "USD" },
-            description: "Open-source MCP server for AI agents with 7 read-only LLM capacity planning tools. Supports stdio and HTTP transports.",
-            version: "0.2.0-alpha.1",
+            description: "Open-source MCP server for AI agents with 7 read-only LLM capacity planning tools. Supports stdio + HTTP + hosted endpoint at tokcalc.vercel.app/api/mcp.",
+            version: "0.2.0",
             license: "https://www.apache.org/licenses/LICENSE-2.0",
             url: "https://tokcalc.vercel.app/mcp",
             downloadUrl: "https://www.npmjs.com/package/@tokcalc/mcp-server",
```

src/app/mcp/route.ts +353

```
@@ -0,0 +1,353 @@
+/**
+ * tokcalc MCP Server — Public hosted endpoint (v0.2.0 final).
+ *
+ * URL: https://tokcalc.vercel.app/api/mcp
+ *
+ * Implements the MCP Streamable HTTP transport (protocol version 2025-03-26)
+ * in STATELESS mode with BEARER API KEY auth + KV-BACKED RATE LIMITING.
+ *
+ * Uses the Web-standard transport (`WebStandardStreamableHTTPServerTransport`)
+ * because Next.js Route Handlers use the Web Fetch API (Request/Response),
+ * not Node's IncomingMessage/ServerResponse.
+ *
+ * ─────────────────────────────────────────────────────────────
+ * Required env vars (set on Vercel: Settings → Environment Variables):
+ *   MCP_API_KEY                — bearer API key clients must send
+ *   UPSTASH_REDIS_REST_URL     — Upstash Redis REST URL (for rate limiting)
+ *   UPSTASH_REDIS_REST_TOKEN   — Upstash Redis REST token
+ *
+ * Optional overrides:
+ *   MCP_RATE_LIMIT_ANON_IP_PER_MIN      (default: 30)
+ *   MCP_RATE_LIMIT_AUTHED_KEY_PER_MIN   (default: 120)
+ * ─────────────────────────────────────────────────────────────
+ *
+ * Test with curl (after deploy):
+ *   curl -X POST https://tokcalc.vercel.app/api/mcp \
+ *     -H "Content-Type: application/json" \
+ *     -H "Accept: application/json, text/event-stream" \
+ *     -H "MCP-Protocol-Version: 2025-03-26" \
+ *     -H "Authorization: Bearer $MCP_API_KEY" \
+ *     -d '{"jsonrpc":"2.0","id":1,"method":"tools/list","params":{}}'
+ *
+ * Cursor config (for hosted HTTP):
+ *   {
+ *     "mcpServers": {
+ *       "tokcalc": {
+ *         "url": "https://tokcalc.vercel.app/api/mcp",
+ *         "headers": { "Authorization": "Bearer <your-api-key>" }
+ *       }
+ *     }
+ *   }
+ *
+ * Claude Desktop config (for hosted HTTP — newer Claude versions):
+ *   {
+ *     "mcpServers": {
+ *       "tokcalc": {
+ *         "type": "http",
+ *         "url": "https://tokcalc.vercel.app/api/mcp",
+ *         "headers": { "Authorization": "Bearer <your-api-key>" }
+ *       }
+ *     }
+ *   }
+ */
+
+import { NextRequest } from "next/server";
+import { WebStandardStreamableHTTPServerTransport } from "@modelcontextprotocol/sdk/server/webStandardStreamableHttp.js";
+import { createMcpServer } from "@tokcalc/mcp-server/server.js";
+import { Ratelimit } from "@upstash/ratelimit";
+import { Redis } from "@upstash/redis";
+import crypto from "node:crypto";
+
+// Force Node.js runtime (we need crypto module + Upstash client works on Node)
+export const runtime = "nodejs";
+// Never cache MCP responses — each request needs fresh transport + server
+export const dynamic = "force-dynamic";
+// Allow up to 30s for tool calls (some calculate() calls are compute-heavy)
+export const maxDuration = 30;
+
+// ============================================================
+// CONSTANTS
+// ============================================================
+
+const ANON_IP_LIMIT = Number(process.env.MCP_RATE_LIMIT_ANON_IP_PER_MIN ?? 30);
+const AUTHED_KEY_LIMIT = Number(process.env.MCP_RATE_LIMIT_AUTHED_KEY_PER_MIN ?? 120);
+const MAX_BODY_BYTES = 256 * 1024; // 256KB
+
+// ============================================================
+// SINGLETONS (lazy-initialized)
+// ============================================================
+
+let redisClient: Redis | null = null;
+let anonIpLimiter: Ratelimit | null = null;
+let authedKeyLimiter: Ratelimit | null = null;
+
+function getLimiters(): { anonIp: Ratelimit; authedKey: Ratelimit } | null {
+  const url = process.env.UPSTASH_REDIS_REST_URL;
+  const token = process.env.UPSTASH_REDIS_REST_TOKEN;
+  if (!url || !token) return null;
+
+  if (!redisClient) {
+    redisClient = new Redis({ url, token });
+  }
+  if (!anonIpLimiter) {
+    anonIpLimiter = new Ratelimit({
+      redis: redisClient,
+      limiter: Ratelimit.slidingWindow(ANON_IP_LIMIT, "1 m"),
+      prefix: "tokcalc-mcp:anon-ip",
+    });
+  }
+  if (!authedKeyLimiter) {
+    authedKeyLimiter = new Ratelimit({
+      redis: redisClient,
+      limiter: Ratelimit.slidingWindow(AUTHED_KEY_LIMIT, "1 m"),
+      prefix: "tokcalc-mcp:authed-key",
+    });
+  }
+  return { anonIp: anonIpLimiter, authedKey: authedKeyLimiter };
+}
+
+// ============================================================
+// AUTH + RATE LIMIT HELPERS
+// ============================================================
+
+function isAuthEnabled(): boolean {
+  return !!process.env.MCP_API_KEY;
+}
+
+function isRateLimitEnabled(): boolean {
+  return !!process.env.UPSTASH_REDIS_REST_URL && !!process.env.UPSTASH_REDIS_REST_TOKEN;
+}
+
+function safeEqual(a: string, b: string): boolean {
+  const aa = Buffer.from(a, "utf8");
+  const bb = Buffer.from(b, "utf8");
+  if (aa.length !== bb.length) return false;
+  return crypto.timingSafeEqual(aa, bb);
+}
+
+function hashKey(key: string): string {
+  return crypto.createHash("sha256").update(key).digest("hex").slice(0, 16);
+}
+
+function extractBearer(req: NextRequest): string | null {
+  const authHeader = req.headers.get("authorization");
+  if (!authHeader) return null;
+  const match = /^Bearer\s+(.+)$/i.exec(authHeader.trim());
+  return match ? match[1].trim() : null;
+}
+
+function getClientIp(req: NextRequest): string {
+  // Vercel sets x-forwarded-for with the real client IP first
+  const xff = req.headers.get("x-forwarded-for");
+  if (xff) return xff.split(",")[0].trim();
+  // Cloudflare
+  const cfIp = req.headers.get("cf-connecting-ip");
+  if (cfIp) return cfIp.trim();
+  // Fallback
+  return "unknown";
+}
+
+interface AuthResult {
+  authenticated: boolean;
+  rateLimitId: string;
+  reason?: string;
+}
+
+function authenticateRequest(req: NextRequest): AuthResult {
+  const expectedKey = process.env.MCP_API_KEY;
+  const clientIp = getClientIp(req);
+
+  // Anonymous mode (MCP_API_KEY not set) — allow access, rate-limit per IP
+  if (!expectedKey) {
+    return {
+      authenticated: false,
+      rateLimitId: `anon-ip:${clientIp}`,
+    };
+  }
+
+  // Authenticated mode — require bearer token
+  const providedKey = extractBearer(req);
+  if (!providedKey) {
+    return {
+      authenticated: false,
+      reason: "Missing Authorization header. Expected: Authorization: Bearer <your-api-key>",
+      rateLimitId: `anon-ip:${clientIp}`,
+    };
+  }
+
+  if (!safeEqual(providedKey, expectedKey)) {
+    return {
+      authenticated: false,
+      reason: "Invalid API key.",
+      rateLimitId: `bad-key:${hashKey(providedKey)}`,
+    };
+  }
+
+  return {
+    authenticated: true,
+    rateLimitId: `key:${hashKey(providedKey)}`,
+  };
+}
+
+interface RateLimitResult {
+  success: boolean;
+  limit: number;
+  remaining: number;
+  reset: number;
+  bucket: "anon-ip" | "authed-key" | "none";
+}
+
+async function checkRateLimit(rateLimitId: string, authenticated: boolean): Promise<RateLimitResult> {
+  const limiters = getLimiters();
+  if (!limiters) {
+    return { success: true, limit: Infinity, remaining: Infinity, reset: 0, bucket: "none" };
+  }
+
+  // Always check per-IP bucket
+  const ipResult = await limiters.anonIp.limit(rateLimitId);
+
+  // If authenticated, also check per-key bucket (looser)
+  if (authenticated && rateLimitId.startsWith("key:")) {
+    const keyResult = await limiters.authedKey.limit(rateLimitId);
+    if (!keyResult.success) {
+      return {
+        success: false,
+        limit: AUTHED_KEY_LIMIT,
+        remaining: keyResult.remaining,
+        reset: keyResult.reset,
+        bucket: "authed-key",
+      };
+    }
+  }
+
+  return {
+    success: ipResult.success,
+    limit: ANON_IP_LIMIT,
+    remaining: ipResult.remaining,
+    reset: ipResult.reset,
+    bucket: "anon-ip",
+  };
+}
+
+// ============================================================
+// ERROR RESPONSE HELPERS
+// ============================================================
+
+function mcpError(statusCode: number, code: number, message: string, detail?: string, extraHeaders?: Record<string, string>): Response {
+  return new Response(
+    JSON.stringify({
+      jsonrpc: "2.0",
+      error: {
+        code,
+        message,
+        data: detail ? { detail } : undefined,
+      },
+      id: null,
+    }),
+    {
+      status: statusCode,
+      headers: {
+        "Content-Type": "application/json",
+        ...extraHeaders,
+      },
+    },
+  );
+}
+
+function unauthorized(reason: string): Response {
+  return mcpError(401, -32001, "Unauthorized", reason, {
+    "WWW-Authenticate": `Bearer realm="tokcalc-mcp"`,
+  });
+}
+
+function tooManyRequests(result: RateLimitResult): Response {
+  const retryAfterSec = Math.max(1, Math.ceil((result.reset * 1000 - Date.now()) / 1000));
+  return mcpError(429, -32002, "Too Many Requests",
+    `Rate limit exceeded on bucket "${result.bucket}". Try again in ${retryAfterSec} second${retryAfterSec === 1 ? "" : "s"}.`,
+    {
+      "Retry-After": String(retryAfterSec),
+      "RateLimit-Limit": String(result.limit === Infinity ? 0 : result.limit),
+      "RateLimit-Remaining": String(result.remaining === Infinity ? 0 : result.remaining),
+      "RateLimit-Reset": String(result.reset),
+      "X-RateLimit-Bucket": result.bucket,
+    },
+  );
+}
+
+// ============================================================
+// POST HANDLER (the main MCP endpoint)
+// ============================================================
+
+export async function POST(req: NextRequest): Promise<Response> {
+  // Validate MCP-Protocol-Version header
+  if (!req.headers.get("mcp-protocol-version")) {
+    return mcpError(400, -32600, "Missing MCP-Protocol-Version header",
+      "Streamable HTTP requires the MCP-Protocol-Version header. Use '2025-03-26' for the latest Streamable HTTP spec.");
```